BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

PHYSICS (PART-I) — (Fresh/Re-appear) — (Version-A) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 18 | **Paper Code:** BN-XI-XXIII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. How many significant figures does 1.362+25.2 have?

(A) 2
(B) 3
(C) 5
(D) 8

Q2. If the dot product of two non-zero vectors A and B is zero, then magnitude of their cross product is

(A) 0
(B) 1
(C) AB
(D) -AB

Q3. In an isothermal change, internal energy

(A) Decreases
(B) Increases
(C) Becomes zero
(D) Remains contant

Q4. The time rate of change of momentum gives

(A) Acceleration
(B) Impulse
(C) Power
(D) Force

Q5. Which of the following forces are conservative?

(A) Electric force
(B) Elastic spring force
(C) Gravitation force
(D) All of them

Q6. Which of the following factors has no effect on the speed of sound in gas?

(A) Humidity
(B) Pressure
(C) Temperature
(D) Density

Q7. If the period of oscillation of mass M suspended from a spring is 2s, then the period of mass 4M will be,

(A) 1s
(B) 2s
(C) 3s
(D) 4s

Q8. At terminal velocity the acceleration of a falling object is

(A) $0ms^{-1}$
(B) $1ms^{-2}$
(C) $-9.8ms^{-2}$
(D) $+9.8ms^{-2}$

Q9. Linear acceleration $a=r\alpha$ when 'θ' is

(A) 0°
(B) 90°
(C) 180°
(D) 360°

Q10. The angle formed by an arc at the center of the circle whose length is equal to the circumference of circle is

(A) 0
(B) $\pi/2$
(C) π
(D) 2π

Q11. If the amplitude of a wave is doubled, then its intensity is

(A) Doubled
(B) Halved
(C) Quadrupled
(D) One fourth

Q12. Which one is constant for a satellite in orbit?

(A) Velocity
(B) K.E
(C) Potential energy
(D) Angular momentum

Q13. Triple point of water is

(A) 273.16°F
(B) 372.16K
(C) 273.16K
(D) 373.16°F

Q14. When Fx=3N and Fy=4N, then F=_____

(A) 7N
(B) 5N
(C) 3N
(D) 4N

Q15. The numerical value of escape velocity on the surface of earth is

(A) $11.2\times10^{3}ms^{-1}$
(B) $11.2\times10^{3}Kms^{-1}$
(C) $11.2ms^{-1}$
(D) $11.2\times10^{3}ms^{-1}$

Q16. SI unit of impulse is;

(A) $Kgms^{-2}$ (B) Ns

(C) Ns^{-1} (D) Nm

Q17. Work is said to be negative when F and d are;

(A) Parallel (B) Anti-parallel

(C) Perpendicular (D) At 45°

Q18. Venturi meter is a device use to measure the

(A) Mass of fluid (B) Viscosity of fluid

(C) Speed of fluid (D) Density of fluid

PHYSICS (PART-I) — (Fresh/Re-appear) — SECTION –B

Time Allowed: 2:40Hrs | **Maximum Marks:** 40

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 40) Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. If an equation is dimensionally correct, is that equation is right equation?

ii. For which angle the equation $|\vec{A}.\vec{B}| = |\vec{A}\times\vec{B}|$ is correct?

iii. Define the following terms, distance, displacement, velocity, acceleration.

iv. Define power and show that power is the dot product of force and velocity.

v. State and explain Stokes Law.

vi. What will be the frequency of a simple pendulum if its length is 1m?

vii. Write conditions for interference.

viii. Find the efficiency of a Carnot's heat engine working between the steam and ice points?

ix. Why does a sound wave travel faster in sloid that in gases?

x. Show that $v = r\omega$

xi. Differentiate between systematic and random errors.

xii. What is linear momentum? State law of conservation of linear momentum.

xiii. Is K.E a vector quantity?

SECTION –C

Maximum Marks: 27

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) Derive mathematically the equation of continuity and relate it to the time rate of volume flow.

(b) A mass at the end of a spring vibrates up and down with a frequency of 0.600Hz and an amplitude of 5cm. What is its displacement 2.56s after it reaches a maximum?

Q4.

a) Define and explain dot and cross product of two vectors. Give examples.

(b) The magnitude of dot and cross product of two vectors $6\sqrt{3}$ and 6 respectively. Find the angle between the vectors.

Q5.

a) Derive a relation between molar specific heat at constant pressure and molar specific heat at constant volume.

(b) The speed of sound in air at 0°C is $332 ms^{-1}$. What will be the speed of sound at 22°C?

Q6. Write notes on any two of the following;

i. Simple pendulum

ii. Newton's 2nd Law

iii. Significant figures